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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/643,401	12/08/2021	Daniel BEDAU	WDA-5852*B-US	5618

131616 7590 06/24/2025

Jacobsen IP Law - WD
1821 S. Bascom Ave. #151
Campbell, CA 95008

EXAMINER
BALL, JOHN C

ART UNIT	PAPER NUMBER
1795	

NOTIFICATION DATE	DELIVERY MODE
06/24/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eofficeaction@appcoll.com
krista@jacobseniplaw.com

Office Action Summary	Application No. 17/643,401	Applicant(s) BEDAU et al.	
	Examiner JOHN C BALL	Art Unit 1795	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 12 June 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 2-17 and 19-23 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☑ Claim(s) 19-23 is/are allowed.

7) ☑ Claim(s) 2-3,8 and 14 is/are rejected.

8) ☑ Claim(s) 4-7,9-13 and 15-17 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED CORRESPONDENCE

Summary

1. This Office Correspondence is based on the Amendment and Reply filed with the Office on 12 June 2025, regarding the Western Digital Technologies, Inc. application.

2. Claims 2-17 and 19-23 are currently pending and have been fully considered.

3. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

4. As the instant claims have already been considered on the merits, this Office Correspondence is made a Non-Final Rejection.

Claim Rejections - 35 USC § 102

5. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

6. Claims 2, 3, 8, and 14 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by a US Patent Application Publication to Marziali, et al. (US 2013/0344498 A1; hereinafter, “Marziali”).

Regarding claim 3, Marziali discloses an invention providing methods for causing motion of particles in a medium ([0020]; which read upon the recited, "[a] system for controlling movement of at least one molecule in a first fluid having a first viscosity"). Marziali teaches a 2D planar surface of a substrate, which supports the medium ([0167]; which reads on "a fluid region defined by at least one fluid-retaining surface, wherein the fluid region is traversable by the at least one molecule"). Marziali also teaches the mobility of particles may be altered by applying a magnetic field to the medium through which particles are passing due to alteration of a viscosity of the medium which may comprise a suitable magneto-rheological fluid ([0104], [0108]; which reads on "a field-responsive fluid situated in the fluid region; and a [inherent] field generator for generating a magnetic ... field across the fluid region, wherein, in response to a magnitude of the magnetic ... field across the fluid region exceeding a threshold magnitude, a viscosity of the field-responsive fluid is greater than the first viscosity"). Additionally, Mariali teaches a voltage source (Figure 3A).

Regarding claim 2, Marziali teaches a 2D planar surface of a substrate, which supports the medium ([0167]).

Regarding claim 8, Marziali teaches the mobility of particles may be altered by applying a magnetic field to the medium through which particles are

passing due to alteration of a viscosity of the medium which may comprise a suitable magneto-rheological fluid ([0104], [0108]).

Regarding claim 14, Marziali teaches particles may be DNA, a biological polymer ([0008]).

Allowable Subject Matter

7. Claims 19-23 are allowed.

8. Claims 4-7, 9-13 and 15-17 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

9. The following is a statement of reasons for the indication of allowable subject matter: The cited Marziali reference is the closest prior art reference to the instant claims. However, Marziali does not anticipate nor render obvious the limitations of the instant indicated claims.

Response to Arguments

10. Applicant's arguments, filed 12 June 2025, with respect to rejections under 35 USC 112(b) have been fully considered and are persuasive. The rejections under 35 USC 112(b) has been withdrawn.

11. Applicant's arguments, filed 12 June 2025, with respect to double patenting have been fully considered and are persuasive. The obvious-type double patenting rejection has been withdrawn.

12. The Applicant did not offer argument to the rejections under 35 USC 102(a)(1), rather the claims were amended to incorporate into now independent claim 3 formally indicated allowable subject matter. However, after further consideration of the teachings of the prior art, claim 3 is found to be anticipated by the prior art, as outlined above.

Interview with the Examiner

13. If at any point during the prosecution it is believe an interview with the Examiner would further the prosecution of an application, please consider this option. The Automated Interview Request form (AIR) is available to request an interview to be scheduled with the Examiner. First, an authorization for internet communications regarding the case should be filed prior or with an AIR online request.

The internet communication authorization form (SB/0439), which authorizes or withdraws authorization for internet-based communication (*e.g.*, video conferencing, email, etc.) for the application must be signed by the applicant or the attorney/agent for applicant. The form can be found at: https://www.uspto.gov/sites/default/files/documents/sb0439.pdf

The AIR form can be filled out online, and is automatically forwarded to the Examiner, who will call to confirm a requested time and date, or set up a mutually convenient time for the interview. The form can be found at: https://www.uspto.gov/patent/uspto-automated-interview-request-air-form.html

The Examiner encourages, but does not require, interviews by the USPTO Microsoft Teams video conferencing. This system allows for file-sharing along audio conferencing. Microsoft Teams can be used as an internet browser add-on in Microsoft IE, Google Chrome, or Mozilla Foxfire, or as a temporary Java-based application on these browsers. Steps for joining an Examiner setup Microsoft Teams can be found at the USPTO website:
https://www.uspto.gov/patents/laws/interview-practice#step3
Additionally, a blank email to the Examiner at the time of a telephonic interview can be used for a reply to easily allow for Microsoft Teams communication.
Please note, policy guidelines regarding Internet communications are detailed at MPEP §500-502.3, and office policy regarding interviews are detailed at MPEP §713.

Conclusion

14. Any inquiry concerning this communication or earlier communications from the examiner should be directed to JOHN C BALL whose telephone number is (571)270-5119. The examiner can normally be reached M - F, 9 am - 5:30 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule

an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Luan Van can be reached at (571)272-8521. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/J. Christopher Ball/
Primary Examiner, Art Unit 1795
20 June 2025